

Lem aprox fn simple

Lema 1. Sea (X, Σ, μ) un espacio de medida y $f: X \rightarrow [0, \mathbb{R}]$ medible

$$\implies \exists (s_n)_{n \in \mathbb{N}} \subset \mathcal{S} : \forall x \in X : 0 \leq s_n(x) \leq s_{n+1}(x) \leq f(x) \wedge \lim_{n \rightarrow \infty} s_n(x) = f(x).$$

Demostración: Descomponiendo “en filas” la imagen de f , tenemos que la siguiente sucesión cumple las condiciones de la proposición:

$$\forall n \in \mathbb{N} : s_n := \sum_{m=0}^{n2^n} \frac{m}{2^n} \mathbb{1}_{E_{n,m}} \text{ con } E_{n,m} := \left\{ x \in X : \min(f(x), n) \in \left[\frac{m}{2^n}, \frac{m+1}{2^n} \right) \right\}$$

Se tiene que $\forall n \in \mathbb{N} : s_n$ es simple por ser suma finita de funciones características escaladas. Veamos que es creciente. Sea $x \in X$ y $y = f(x)$.

$$- y = \infty \implies s_n = n \implies \forall n \in \mathbb{N} : s_n \leq s_{n+1}.$$

$$- y < \infty \implies s_n(x) = \frac{\lfloor n2^n \rfloor}{2^n} \text{ y queremos ver que } \frac{\lfloor (n+1)2^{n+1} \rfloor}{2^{n+1}} \geq \frac{\lfloor n2^n \rfloor}{2^n}.$$

$$\begin{aligned} & \frac{\lfloor (n+1)2^{n+1} \rfloor + 1}{2^{n+1}} > y \geq \frac{\lfloor n2^n \rfloor}{2^n} \implies \frac{\lfloor (n+1)2^{n+1} \rfloor + 1}{2^{n+1}} > \frac{\lfloor n2^n \rfloor}{2^n} \\ \implies & \lfloor (n+1)2^{n+1} \rfloor + 1 > 2 \lfloor n2^n \rfloor \implies \lfloor (n+1)2^{n+1} \rfloor \geq 2 \lfloor n2^n \rfloor \\ \implies & \frac{\lfloor (n+1)2^{n+1} \rfloor}{2^{n+1}} \geq \frac{2 \lfloor n2^n \rfloor}{2^{n+1}} = \frac{\lfloor n2^n \rfloor}{2^n} \implies s_{n+1}(x) \geq s_n(x) \implies s_{n+1} \geq s_n. \end{aligned}$$

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